
AI AGENTS

WORKSHOP SERIES

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UCLA MQE Spring 2026

SCAN THE QR CODE TO SIGN IN



Access to Materials: <https://github.com/lorayc/ai-agents>

THE WORKSHOP SERIES

01

Foundations

API calls, tool use, ReAct loop, FRED agent

02

RAG & Memory

Embeddings, vector stores, FOMC document search

03

Multi-Agent Systems

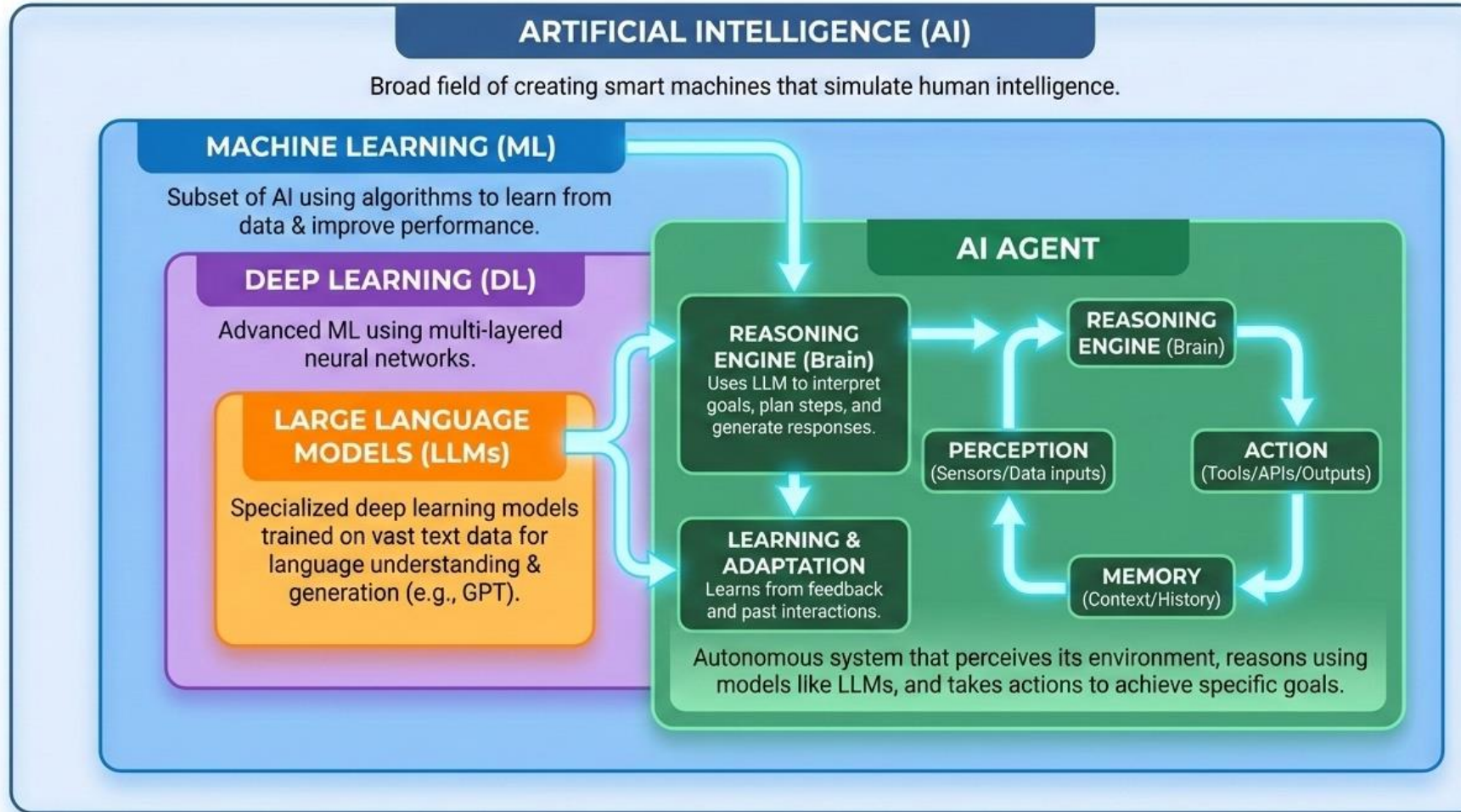
Orchestration, delegation, parallel research

04

Production & Eval

Guardrails, evaluation, deployment patterns

THE ECOSYSTEM



AI AGENT

Agent = LLM + Memory + Tools + Policy + Objective



Perception

Observe documents, APIs, databases, and external state.



Reasoning

Inference and decomposition over the gathered information.



Action

Execute tools, code, searches, calculations, API calls.



Memory

Retain intermediate state, prior outputs, long-term knowledge.



Objective

Explicit or implicit optimization target.

SINGLE VS MULTI-AGENT ORCHESTRATION

SINGLE - A G E N T

User → LLM → Output

STRENGTHS

- ✓ Simple architecture
- ✓ Cheap to operate
- ✓ Low latency

LIMITS

- ✗ Context overload
- ✗ Hallucination accumulation
- ✗ Weak decomposition
- ✗ Poor long-horizon reasoning

M U L T I - A G E N T

Coordinator → specialists → aggregator

Research

Analyst

Coder

Critic

Writer

Risk

ORCHESTRATION PATTERNS

Hierarchical

Supervisor → specialists → aggregation

Most common in production. Interpretable and controllable but creates bottlenecks at the coordinator.

Swarm / Decentralized

Agents interact peer-to-peer

Robust and emergent, but unstable coordination and exploding communication costs.

Blackboard

Shared memory environment

Agents read from and write to a common workspace. Transparent state, but contention and noise.

Planner–Executor

Planner decomposes, executors run

A planner builds a task queue; executors process it. Strong for coding, research, workflows.

MEMORY

SHORT-TERM

Context window

Lives in the prompt. When problem complexity exceeds context length, you compress — summarize, retrieve, restate.

LONG-TERM

External stores

Vector stores, knowledge graphs, document stores. Pipeline: query → embedding → vector search → relevant chunks → agent context.

SHARED vs PRIVATE

A core trade-off

Shared memory enables coordination and transparency at higher cost; private memory cuts noise and is efficient. Mirrors public vs. private information.

ROUTING

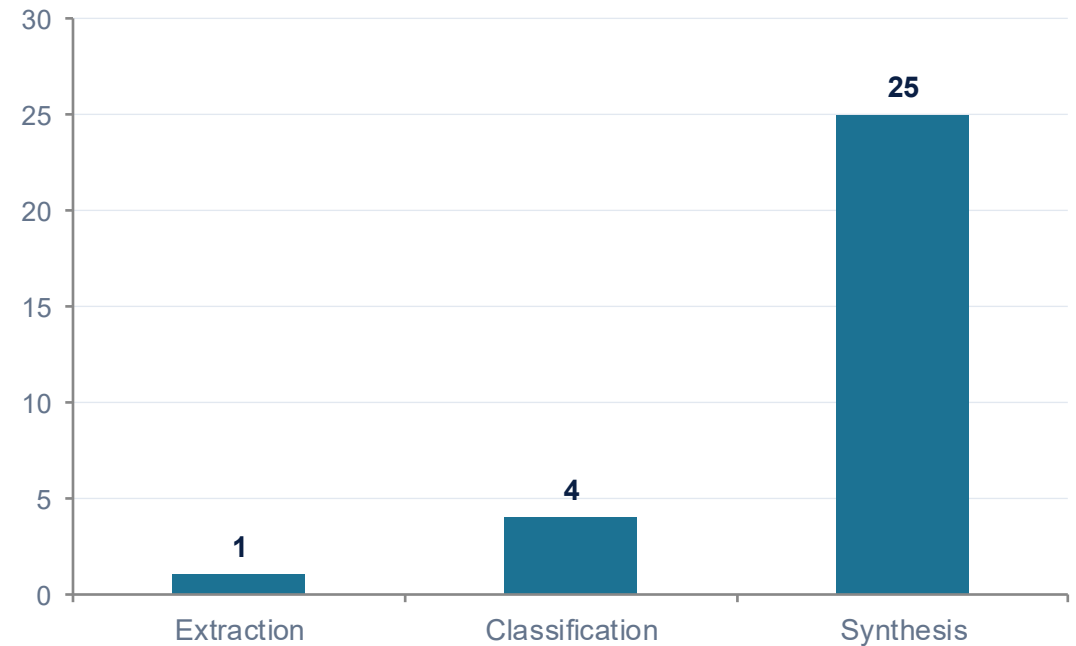
Large systems fail when every agent sees everything.

A controller decides: $P(\text{route to agent}_i | x)$

THIS IS

- Mixture-of-experts in disguise
- Conditional computation
- Adaptive resource allocation
- Optimal labor allocation, in economic terms

COST-AWARE ROUTING EXAMPLE



COST EFFICIENCY

Total Cost = Σ price $\times$ tokens, summed across every agent in the system.

1×

EXTRACTION

Cheap small model. High volume,
narrow task.

4×

CLASSIFICATION

Mid-tier model. Pattern recognition
over structured text.

25×

SYNTHESIS

Frontier model. Long-context
reasoning, novel composition.

AGGREGATING BELIEFS ACROSS AGENTS

AGGREGATION MECHANISMS

- 1 Voting**
Each agent casts a vote, majority wins.
- 2 Weighted averaging**
Confidence-scaled estimates.
- 3 Debate loops**
Thesis → critique → defense → judge.
- 4 Critique cycles**
Generate → critique → revise.
- 5 Confidence scoring**
Self-reported certainty drives weights.

EXAMPLE: DEBATE PIPELINE

Agent A States the thesis



Agent B Issues critique



Agent A Defends position



Judge Evaluates and decides

APPLICATIONS



Central bank monitoring

Parse FOMC minutes, extract inflation tone, classify labor conditions, build diffusion indexes.



Financial market intelligence

Monitor earnings calls, macro releases, news flows, prediction markets in near real-time.



Macro forecasting

Specialized agents estimate inflation, growth, labor, and stress; aggregate into ensemble forecasts.



Policy simulation

Simulate households, firms, regulators, central banks. Emergent macro behavior from local rules.

AGENTS FOR FINANCIAL SERVICES

CLAUDE MANAGED AGENTS

- Faster, scalable deployment of autonomous agents

AGENTS FOR FINANCIAL SERVICES

- Ready-to-run agent templates
- Building pitchbooks, closing the books at month-end, etc.
- Plugin or managed agents in Claude

	Messages API	Claude Managed Agents
What it is	Direct model prompting access	Pre-built, configurable agent harness that runs in managed infrastructure
Best for	Custom agent loops and fine-grained control	Long-running tasks and asynchronous work
Learn more	Messages API docs	Claude Managed Agents docs

The screenshot displays the ARQOS Valuation Reviewer interface. At the top, there's a navigation bar with 'Funds', 'Investors', 'Agents', and 'Reports'. The main header shows 'Valuation Reviewer' for 'Fund Prudent & Wise Growth IV' and 'GP Prudent & Wise Capital' for 'Period Q1 2028'. A 'Run Q1 2028 close' button is visible. Below the header, a 'WORKFLOW' section shows steps: 'Ingest', 'Value', 'Review', and 'Distribute'. A message states 'Q1 packages received for 8 portfolio companies - GP Portal (5) - Email - fund-ops@arqos (3) - as of Mar 31, 2028'. The main area displays a grid of 8 portfolio company cards, each with a 'QUEUED' status and a list of documents:

- Greater Midwestern Bolt & Flange** (Industrial Manufacturing): Q1-26 Financials.pdf, Cap Table Q1-26.xlsx, Debt Schedule.pdf
- Two Rivers Cartage** (Transportation & Logistics): Q1-26 Pack.pdf, Fleet Schedule.xlsx, Covenants.pdf
- Tri-County Imaging & Wellness** (Healthcare Services): Financials Q1.pdf, Payroll Mtx.xlsx, Mgmt Update.docx
- Prairie Door & Window** (Building Products): Q1 Reporting Pack.pdf, Backlog.xlsx
- Ledger & Quill Software** (Vertical SaaS): ARR Snapshot.pdf, Q1 P&L.xlsx, Board Deck.pdf
- Honest Pretzel Co.** (Consumer Packaged Goods): Q1 Pack.pdf, Trade Spend.xlsx, Debt Schedule.pdf
- Bicuspid Partners** (Multi-site Healthcare): Q1 Financials.pdf, Same-store.xlsx, MSA Pipeline.docx
- Permian Mud & Pipe** (Energy Services): Q1 Pack.pdf, Station.xlsx, Book.pdf

 A tooltip at the bottom right reads: 'New agents for financial services | Claude | Cowork + Claude Managed Agents'.

BEIGE BOOKS APPLICATION

OVERVIEW

Architecture

The orchestrator, deterministic tools, model-backed agents, database cache, and export layer.

Process

Trace a report from PDF discovery to evidence extraction, category scoring, validation, indexing, and export.

Decision Choices

Why the system uses strict JSON, Pydantic schemas, cache keys, prompt versions, offline mode.

Tradeoffs

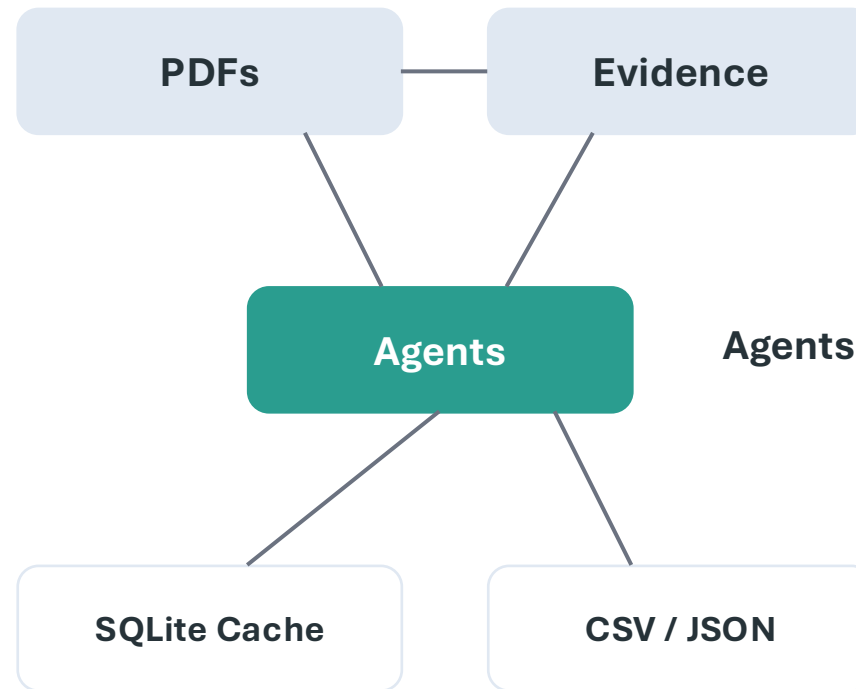
Reliability, cost, repeatability, and research quality across LLM-backed and deterministic paths.

ORCHESTRATION

Multi-agent pipeline

GOAL: Create an economic sentiment index from the Federal Reserve Beige Books.

Input: PDFs named
BeigeBook_YYYYMMDD.pdf

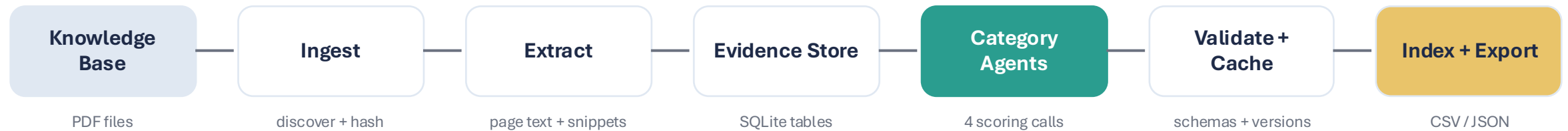


Evidence: candidate snippets selected by
category keywords and page location

Agents: one scoring task per economic category

Output: component scores plus an equal-weight aggregate index

ARCHITECTURE FLOW



- Orchestration & decomposition
- NLP and sentiment analysis
- Macroeconomic interpretation
- Retrieval pipelines
- Cost optimization across model sizes
- Ensemble reasoning

PROCESSING STEPS

1. Register reports

Find matching PDFs, parse dates, compute file hashes, and upsert report metadata.

2. Extract or reuse evidence

If the current extraction version is cached, reuse it. Otherwise extract pages and snippets.

3. Score each category

For each configured category, skip cached scores or call the selected scoring agent.

4. Validate output

Parse JSON, repair invalid JSON if needed, and enforce the CategorySentiment schema.

5. Persist decisions

Store score, confidence, rationale, evidence, prompt version, and model.

6. Build index

Require all component scores, average them equally, then export CSV and JSON.

CATEGORY AGENTS

Labor Market

Hiring, unemployment, wages, labor availability, layoffs, hours.

Prices / Inflation

Input costs, selling prices, inflation pressure, pass-through, margins.

Consumer Activity

Retail, travel, leisure, household spending, credit-sensitive demand

Business Conditions

Manufacturing, services, lending, investment, outlook, firm activity.

PROMPT AND OUTPUT CONTRACT

-2**strong contraction****-1****weakening****0****mixed / little
changed****1****improving****2****strong expansion**

Prompt contains

Report date, category label, category definition, exact integer score scale, JSON shape, and evidence excerpts only.

Output must include

Category, score, confidence, one-sentence rationale, and cited evidence strings. Invalid JSON can be repaired, then validated.

OUTPUT

Report date	Aggregate	Labor	Prices	Consumer	Business
2026-01-14	0.00	-1	1	0	0
2026-03-04	0.00	-1	1	0	0
2026-04-15	-0.25	0	-1	0	0

FORMULA

aggregate_score = average(labor_market, prices_inflation, consumer_activity, business_conditions)

A transparent. All categories currently receive equal weight.

SUMMARY

Multi-Agent Orchestration =

Economic Coordination + Distributed Computing
+ Probabilistic Reasoning + LLMs

A multi-agent system is a computational economy of specialized reasoning entities coordinating under constraints to solve complex tasks.